BL-32 Demonstration and Handover

Backlog item BL-32 (issue #32), NFR-09. Prepared 8 November 2026; updated 9 November 2026 for the redesigned interface, timetable, semester classes and emailed login details.

This guide is for anyone who needs to run, demonstrate or assess the Student Attendance Management System: evaluators, new team members and the Admin who takes it over. It covers how to prepare a clean demo, a tested demonstration script, how each role uses the system, which requirements are complete, where the test evidence is, and what is still limited.

The demonstration script in section 3 was rerun on 9 November 2026 against a freshly migrated database loaded with the demo data; all 15 checks passed. Alert emails only reach a student who has a real email address on their account (see section 3, step 4).

1. Prepare a clean demo environment

Follow the full Setup section of the README the first time. In short:

1. Install Python 3.12+, PostgreSQL 16+ and Git, then clone the repository and install the dependencies into a virtual environment:
   ```
   python3 -m venv .venv
   source .venv/bin/activate            # Windows: .venv\Scripts\activate
   pip install -r requirements-dev.txt
   ```
2. Create a **new, empty** PostgreSQL database for the demo. The demo data refuses to load into a database that already has it.
3. Copy `.env.example` to `.env` and set:
   - `SECRET_KEY`: generate one with the command shown in `.env.example`.
   - `DATABASE_URL`: points at the demo database.
   - `DEBUG=True`: the demo data only loads in development mode.
   - `BACKUP_ENCRYPTION_KEY`: generate one with the command shown in `.env.example`. Without it, backups are switched off.
4. Create the tables, load the demo data and start the server:
   ```
   python manage.py migrate
   python manage.py seed_test_data
   python manage.py runserver
   ```
5. Open http://127.0.0.1:8000/login/.

Two-factor login. It is on by default, so each account sets up an authenticator app (Google Authenticator or Microsoft Authenticator) at first login and saves its 10 backup codes. For a quicker walkthrough on your own machine only, add `OTP_REQUIRED=False` to `.env` and restart the server. Show two-factor login at least once with it switched on (step 6 of the script).

Email. By default, emails are printed in the terminal running the server; none are delivered. To send real email through Gmail, run `python scripts/configure_gmail.py` (Windows: `scripts\Configure-Gmail.cmd`) and restart the server. The demo accounts have **no email addresses**, so give any account you want to receive email a real address first, using **Students -> View -> Edit student** (or **Faculty -> View -> Edit**).

2. Demo accounts

All demo accounts use the password `Attend#2026`. The data is fictional; see Test_Data_Fixture.md for every figure.

Role	Username	What it shows
Admin	`ADMIN1`	Everything
Faculty	`F001`	Teaches CS101 Programming in Section A (S001, S002, S004); timetabled Monday period 2, Wednesday period 1, Friday period 3
Faculty	`F002`	Teaches MA101 Mathematics in Section B (S001, S003); timetabled Monday period 4, Tuesday period 2, Thursday period 5
Faculty	`F003`	Has no classes, so the dashboard shows zeros
Student	`S001`	Below 75% in both subjects and overall: 76/103 (73.79%). Semester classes: CS101 120, MA101 40
Student	`S002`	Exactly 75.00%, so not flagged (equal is not below)
Student	`S003`	No classes held yet: "N/A - no classes yet"
Student	`S004`	76.00%, above the threshold

3. Demonstration script (about 15 minutes)

Use a normal and a private browser window so two people can be signed in at once. Every page has a sidebar on the left (on a phone, open it with the Menu button); the steps name the sidebar entry to click.

Step	Who	Do this	You should see
1. Today's classes	`F001`	Log in and look at the dashboard.	A banner with today's date and the next class, and **Today's classes** from

Step	Who	Do this	You should see
			the timetable. On a Monday: CS101, Period 2 (9:00-10:00 am); after 10:00 it is flagged **not marked yet**. Saturday and Sunday have no classes.
2. Mark attendance	`F001`	**Mark attendance** next to today's class (or sidebar **Mark attendance** -> CS101, today, the timetabled period -> **Load roster**). Mark S001 Absent, click **Mark remaining as Present** -> **Save complete attendance**.	The saved roster with S001 Absent and S002, S004 Present, marked "already saved". A second submission with different marks is refused. Future dates are greyed out. The class shows **Marked** on the dashboard.
3. Correct a mark	`F001`	Sidebar **History and corrections** -> **Apply filters** -> find today's S001 record -> **Correct** -> Present, reason "Arrived late" -> **Save correction** -> **View changes**.	The record is Present. The history shows old/new status, who changed it, when, and the reason.
4. Low-attendance alert	`S001`	Sidebar **Notifications**.	**Where you stand now**: CS101 75/101 (74.26%), MA101 2/3 (66.67%) and overall 77/104 (74.04%), each with the classes needed, e.g. "Attend at least 15 of the 19 classes left this semester". Below it, the alert created when step 2 saved the absence ("attend the next 7 classes to get back to 75%"). The email copy is printed in the server terminal, or delivered if Gmail is set up **and** S001 has a real email address.
5. Student view	`S001`	**Dashboard**, then click the **CS101** card. Back on **My attendance**, filter MA101 from 2 to 3 October 2026.	Overall attendance and **Subjects below threshold: 2**, then one card per subject with its percentage. The CS101 page lists every class with date, time and Present/Absent (20 per page). The filter shows 1/2 (50.00%).
6. Reports	`F001`	Sidebar **Reports** -> today to today, CS101 -> **Generate report** -> **Download CSV** and **Download PDF**.	A three-student report and two downloaded files. Faculty only see their own classes.
7. Timetable and semester classes	`ADMIN1`	Sidebar **Timetable**; try adding MA101 to Section B on Monday period 4. Then **Semester classes**.	Six weekly classes by day. The duplicate is refused and the clash is named. Semester totals CS101 120 and MA101 40.
8. Send login details	`ADMIN1`	Sidebar **Accounts and resets** -> **Reset password** for S004 -> type an email address under **Send these details by email** -> **Send**.	"Login details sent to s***@..." (or printed in the terminal without Gmail). The address is saved on S004's account.
9. Two-factor login	any	Log out and log in again with `OTP_REQUIRED=True`.	A QR code at first login, then 10 backup codes; later logins ask for the 6-digit code.
10. Shortages and threshold	`ADMIN1`	Sidebar **Below threshold**, then **Attendance threshold** -> 80 -> **Save threshold**, and open the list again.	At 75%: S001 overall and in both subjects. At 80%: S001, S002 and S004 overall. The dashboard shortage box appears only while someone is below.
11. Backup and recovery	`ADMIN1`	**Backup and recovery** -> **Back up now** -> **Download encrypted file**. Change the threshold to 85, then **Restore...** on that backup, enter your password, tick the confirmation -> **Restore backup**. Log in again.	A completed, encrypted backup and a `.sambak` download. After the restore the threshold is back to 80; everyone is signed out and a safety backup was taken first.
12. Permissions	`F001`, `S001`	Try to open `/students/create/`, `/timetable/` or `/backups/` as Faculty or Student.	Access is refused; nothing changes. The sidebar never shows these pages to them.

Optional extras: **Forgot password?** on the login page (the Admin then sees "Reset requested" in **Accounts and resets**), adding a student or faculty member, and the 15-minute idle sign-out.

Recorded walkthroughs (made before the 9 November redesign) are in demo/sprint-2 and demo/sprint-1.

4. User guide by role

Everyone

- **Log in** with your username (roll number, faculty ID or Admin username, any letter case) and password, then the 6-digit code from your authenticator app. No phone? Use **Email me a code** or one of your backup codes.

- **First login:** set up the authenticator app from the QR code, save the 10 backup codes (**Download as text file** or **Print**), then change your temporary password.
- **Forgot your password?** Use **Forgot password?** on the login page. The Admin is told, resets it, and a new temporary password is emailed to you.
- After 5 wrong passwords or 5 wrong codes, the account pauses for 15 minutes.
- **My profile** shows your own details and the college (PES University), read-only. **Change password** is on the same page.
- The **sidebar** lists only the pages your role can open; your name in the top bar opens **My profile**. **Log out** is at the top right. You are signed out automatically after 15 minutes without activity.

Admin

Task	Where (sidebar)
Add, edit, deactivate, reactivate or permanently delete students (a reason is required to delete)	**Students**
The same for faculty; faculty still teaching a class must be reassigned first	**Faculty**
Create sections ("Dept and sem", "Section", optional year), add subjects with their teacher, enroll students	**Classes and sections**
Create, rename, offer or delete subjects (a subject with attendance cannot be deleted)	**Subjects**
Weekly timetable: add a class by day and period (clashes are refused), remove it	**Timetable**
Classes each subject will hold this semester (drives the students' "classes needed" advice)	**Semester classes**
Reset a password, resend login details, reset someone's two-factor login, handle reset requests; email a temporary password with **Send**	**Accounts and resets**
Add another Admin	**Create Admin account**
Correct any record at any time, with a reason	**History and corrections**
Record a session faculty missed, with a reason	**Record missed attendance**
Reports with CSV/PDF export for any student, class or subject	**Reports**
See who is below the threshold, overall and per subject	**Below threshold**
Change the threshold (default 75%) or the faculty correction window (default 2 days, 0-30)	**Attendance threshold**, **Correction window**
Back up, schedule daily backups, download, import or restore	**Backup and recovery**
Today's classes across the college and which are marked	**Dashboard**

Faculty

Task	Where (sidebar)
Today's classes, the next class and any class not yet marked	**Dashboard**
Mark attendance for an assigned class: every student must be marked; **Mark remaining as Present** fills the rest	**Mark attendance** (or the button beside a class on the dashboard)
See your classes, students and subjects	**My classes**, **Students in my classes**, **My subjects**
Correct a mark within the correction window (until 23:59 IST two days after the session, by default); older changes go to the Admin	**History and corrections** -> **Apply filters** -> **Correct**
Reports and CSV/PDF exports for your own classes	**Reports**
Students below the threshold in your subjects; you also get a Monday 08:00 email when the list is not empty	**Below threshold**

Student

Task	Where (sidebar)
Overall attendance, subjects below the threshold and each subject's percentage	**Dashboard**
Attendance per subject with filters, what you need to reach the threshold, and each class marked Present or Absent (**View classes**)	**My attendance**
Where you stand now, the classes you need, and alerts when a subject or overall figure falls below the threshold (one alert per drop, also by email)	**Notifications**
Your roll number, section, subjects, teachers and college	**My profile**

5. Requirement completion

All backlog items are Done on the project board. BL-32 is this handover; BL-35 to BL-40 record the 9 November additions.

Requirement	Delivered by	Status	Evidence (Test Case Document group)
FR-01 Account provisioning and login	BL-04, BL-34	Complete	UC-01
FR-02 Role management	BL-05	Complete	UC-01
FR-03 Student management	BL-07	Complete	UC-02

Requirement	Delivered by	Status	Evidence (Test Case Document group)
FR-04 Faculty management	BL-08, BL-11	Complete	UC-03
FR-05 Subject management	BL-09	Complete	UC-04
FR-06 Class/section management	BL-10, BL-11	Complete	UC-05
FR-07 Mark attendance	BL-12, BL-13	Complete	UC-06
FR-08 Update attendance	BL-17	Complete	UC-07
FR-09 View attendance	BL-15	Complete	UC-08
FR-10 Attendance history	BL-16, BL-39	Complete	UC-11
FR-11 Attendance calculation	BL-14	Complete	UC-06, UC-08
FR-12 Attendance reports	BL-20	Complete	UC-09
FR-13 Low-attendance detection	BL-22, BL-24	Complete	UC-10
FR-14 Notifications	BL-23	Complete	UC-10
FR-15 Search and filter	BL-19	Complete	UC-11
FR-16 Report export	BL-21	Complete	UC-09
FR-17 Dashboards (and My profile)	BL-25, BL-35, BL-39	Complete	UC-12
FR-18 Modification history	BL-18	Complete	UC-07
FR-19 Backup and recovery	BL-26, BL-27	Complete	UC-13
FR-20 Logout	BL-06	Complete	UC-12
FR-21 Two-factor login	BL-33	Complete	UC-01
FR-22 Timetable and class reminders	BL-36	Complete	UC-14
FR-23 Semester classes and attendance advice	BL-37	Complete	UC-14, UC-08, UC-10
FR-24 Send login details by email	BL-34, BL-38	Complete	UC-01

Requirement	Status	Basis
NFR-01 Performance	Met locally	PT*01-PT*03 passed with 50 users on one machine on 9 Nov; repeat on the real server
NFR-02 to NFR-04 Security, authentication, authorization	Met	BL-05, BL-31; UC-01 and integration tests
NFR-05 Reliability, NFR-10 Data integrity	Met	BL-13 duplicate/partial-save protection; atomic corrections and restore
NFR-06 Availability	Not measured	Needs a month of monitoring on a deployed server (VAL-02)
NFR-07 Usability	Met	BL-30 manual workflow review, the 9 November redesign (BL-35) and automated checks
NFR-08 Scalability	Met locally	PT_04: 1,000 students, 750,000 marks, 50 users
NFR-09 Maintainability	Met	Ruff lint/format, 556 automated tests (9 Nov), 95.10% line and 90.78% branch coverage
NFR-11 Privacy, NFR-14 Auditability	Met	Role-scoped data, owner-only credential email, audit log
NFR-12 Compatibility	Partly verified	Chromium, Firefox and WebKit at 1440, 768 and 320 px (9 Nov); Edge, branded Safari and real phones not tested (VAL-03)
NFR-13 Backup and recovery	Met on macOS	Windows-only file-permission checks not run (VAL-04)
NFR-15 Accuracy	Met	All calculation tests pass

6. Where the evidence is

What	Where
Requirements, design, use cases, test plan, security requirements	Documents/ (SRS v2.1, Software*Architecture*Design(SAD) v1.1, Use*Case*Flow, Test*Plan*Document v2.1, Security*Requirements*Specification)
Changes from the original SRS and design, for the demo	Changes_from_Original_Design.md
Agreed rules and decisions (D-01 to D-28)	BL-01_Decision_Log.md
Part 3 test reports and the 224-case Test Case Document	docs/
Performance, browser and recovery verification records	Documents/Verification/
Implementation notes per backlog item	`Documents/BL-*_Implementation.md` and related notes
Sprint demo videos	demo/
Running the tests	`python manage.py test`, `ruff check .` (README, "Running tests and checks")

7. Known limitations and non-blocking issues

None of these blocks the demonstration.

1. **Availability (NFR-06) is not measured.** It needs a month of monitoring on a real server during 08:00-18:00, Monday-Saturday.

2. **Browser and device coverage is partial.** Edge, branded Safari, the full "latest two versions" matrix and real Android/iOS phones were not tested; phone widths (320 and 360 px) were emulated.
3. **Windows backup checks were not run.** Three file-permission tests only run on Windows. They account for most of the code not yet covered by tests.
4. **Performance was measured on one machine.** Repeat PT*01-PT*04 on the real server; network, HTTPS and email delays were not included.
5. **Email needs local setup.** Each machine needs its own Gmail app password in `.env`. The demo accounts have no email addresses, so alert emails, login-detail emails and emailed login codes only reach someone after a real address is added.
6. **Scheduled jobs need the server's scheduler.** Daily backups (`run_scheduled_backups`, every minute) and the Monday faculty email (`send_weekly_shortage_emails`) only run if cron or Task Scheduler calls them. Setting the time in the app does not install a scheduler.
7. **Password reset goes through the Admin.** There are no self-service reset links (BL-01 D-23), and a temporary password stays valid until it is used or replaced.
8. **Only Present and Absent exist.** "Late" and partial credit were deferred (D-06).
9. **Faculty corrections have a deadline.** After the window (default 2 days), only the Admin can correct or add a session, with a reason.
10. **Restoring a backup signs everyone out** and cancels emailed codes and backup codes; the restore needs a maintenance window. Backups over 256 MiB are refused, and connection poolers in transaction mode are not supported.
11. **The demo data is small.** It is for functional demos; the 1,000-student capacity data is generated separately by the BL-29 verification tools.
12. **Timetable and semester totals are entered by hand.** The Admin enters them; there is no automatic timetable generation or import, and the class reminders appear only on the dashboard (no email or push reminder).